

Theoretical and Experimental Investigations on the Growth of SnS van der Waals Epitaxies on Graphene Buffer Layer

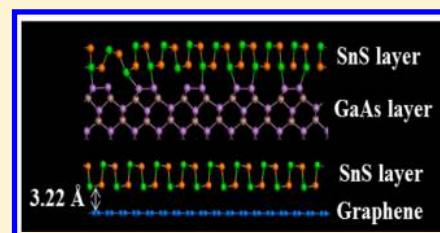
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ABSTRACT: We present theoretical and experimental investigations on the growth of SnS van der Waals epitaxies (vdWEs) on graphene buffer layer (GBL). Local density approximation (LDA) was used to evaluate the bond length disorder, binding energies, and growth orientations for SnS deposited on crystalline semiconductor substrates with and without the GBL. Strong bond length disorder is observed for SnS deposited directly on GaAs substrates, whereas in the case where a GBL is used the disorder is substantially reduced. First-principle calculations indicate two favored growth orientations for SnS deposited on GBL resulting in 12 distinct peaks in the azimuthal hard X-ray diffraction (HXRD) scan due to the structural symmetry of the GBL. The results stipulate formation of strong chemical bonds at the GaAs/SnS interface while the interaction between SnS and the underlying GBL is dominated by vdW force. Nevertheless this vdW force is shown to be strong enough to induce favored nucleation orientations for the SnS and is essential for the observed improvement in the crystallinity of the films.



INTRODUCTION

SnS is a two-dimensional layered semiconductor belonging to the class of material commonly known as vdWEs,¹ which is characterized by strong intralayer covalent bonds and weak interlayer interactions dominated by vdW force. Such unique qualities have led to the belief that vdWEs will have high tolerances for lattice mismatch due to their weak interactions with the substrate.² This unique property may be exploited for the growth of high-quality vdWEs on a wide range of substrates. In addition, vdWE-based heterojunctions can, ideally, be made completely free of interface defects since the bonds within a unit layer are all saturated and interlayer interaction is dominated by vdW force and no chemical bonds are formed across the heterointerface.³ In addition, there are a number of important factors that have led to the increase in the interest in SnS as a potential photovoltaic (PV) material in recent years including (i) earth abundance and nontoxicity of its constituent elements, (ii) a direct bandgap at ~ 1.3 eV and an indirect bandgap at ~ 1 eV, and (iii) high absorption coefficient at 1.5 eV ($>10^4$ cm⁻¹). These properties are especially desirable for PV applications. In particular, the presence of both a direct and indirect bandgap may result in significant enhancement in the carrier lifetime^{4–6} without sacrificing the absorption coefficient. The bandgap for SnS is close to the ideal bandgap size for PV applications and the theoretical conversion efficiency for SnS-based photovoltaic cells (PVCs) is shown to be $>22\%$.⁷

Unfortunately, so far the experimental results on SnS thin films and SnS-based PVCs have demonstrated quite the contrary. Typical grain size is of the order of 0.1 μ m, and the

rocking curve full width at half-maximum (fwhm) is $\sim 3.2^\circ$. This is in stark contrast to the expectation of high tolerance for lattice mismatch for the vdWEs. The highest reported power conversion efficiency for SnS-based heterojunction PV cells is only $\sim 2\%$.^{5,8} Careful examination of the devices reveals at least three possible causes for such poor performance. First and foremost, all devices reported so far exhibit poor crystalline quality for the SnS films with strong dependence of the film quality on the substrates.^{9–11} Such poor film quality partly accounts for the low open circuit voltage, V_{OC} , observed in the devices. Another factor that affects the value of V_{OC} is the poor band alignment for the devices. Work by Reddy in ref 5 indicated a conversion efficiency of 1.3% with a low V_{OC} value of about 0.25 V. In addition to the poor film quality, such low V_{OC} value is also partly attributed to the significant mismatch in the electron affinities (EAs) for SnS and CdS.^{12,13} Recent work by Yue et al.¹⁴ demonstrated significant improvement in the V_{OC} in a SnS nanowire homojunction device to ~ 0.65 V, which is consistent with our analyses above. Third, the doping profile of the p-type absorber layer was not optimized. This will greatly compromise the short circuit current density, J_{SC} , of the device as the inability to establish a back-surface field by varying the doping concentration in the absorber will greatly enhance the recombination of the photogenerated carriers at the back contact.¹⁵

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It is clear from the discussions above that improving the crystal quality and the optimization of the doping profile of the absorber layer are critical factors for the development of high-efficiency SnS-based PVCs. In this paper, we will focus on the theoretical and experimental investigations on the growth of high-quality SnS thin films. So far little work has been done on the theoretical modeling of the growth orientation of SnS. A fundamental understanding of the growth process will greatly facilitate the development of 2D layered material-based devices. Recent work demonstrates that the computation of vdW force using C6 coefficients from local density approximation (LDA) and Perdew–Burke–Ernzerhof wave functions^{16–18} can well describe the dispersion interactions. Some theoretical investigations on the growth of vdWes using LDA and generalized gradient approximation to density functional theory had been performed.^{19–21}

THEORETICAL INVESTIGATION

We have conducted first-principles calculations on the interfacial stability and the orientation of SnS on GaAs(100) (2D/3D system) and SnS on graphene (2D/2D system) using the Vienna *ab initio* simulation package^{22,23} with projector augmented wave potentials.²⁴ The LDA²⁵ was used to describe the exchange–correlation functional in order to properly illustrate the weak interaction. It is noted that vdW functionals often provide more accurate results for dispersion interactions. Nevertheless, LDA is adopted here due to the faster computational speed. Because the system energy is calculated for multiple configurations with different relative orientations of the film with respect to the substrate, a fast approach such as LDA is important. LDA is still widely used for dispersion interactions and, in general, performs better than GGA. More importantly, our main emphasis is to compare the relative system energy among various configurations, rather than extracting any exact individual value of interest. Many systematic errors can be canceled out, and we believe that our results are reasonably justified. The kinetic energy cutoff was chosen to be 450 eV. All structures were optimized by the conjugate gradient method with a convergence criterion, applied to the forces on the atoms, of 0.01 eV/Å.

One of the most important factors with critical impact on the film quality is the formation of chemical bonds at the interface. The SnS/GaAs(100) interface was modeled as a slab with six GaAs layers plus one SnS layer. Additional hydrogen atoms were chosen to saturate the dangling bonds of Ga atoms at the bottom of the slab. The SnS/graphene interface was modeled by placing a SnS layer on a single layer of graphene substrate. A vacuum region of ~12 Å above the film was used to eliminate the unphysical interactions between periodic images. Owing to the large lattice mismatch between SnS and graphene (GaAs) super cells of lateral dimension $4.26 \times 27.06 \text{ Å}^2$ and $39.57 \times 3.96 \text{ Å}^2$ were used for the analyses of the SnS/graphene and the SnS/GaAs(100) interfaces, respectively. Such a treatment ensures the lattice strains of SnS, graphene, and GaAs are smaller than 2.3%. The Brillouin zone was sampled by $10 \times 2 \times 1$ k-points in the case of SnS/graphene and $1 \times 8 \times 1$ k-points for SnS/GaAs(110) respectively. Figure 1 shows the optimized structures of SnS/GaAs(100) and SnS/graphene interfaces. Significant structural distortion of the SnS layer on GaAs(100) substrate is shown in Figure 1a. To quantify the interfacial distortions, we have calculated the bond-disorder factor, ΔX , given by

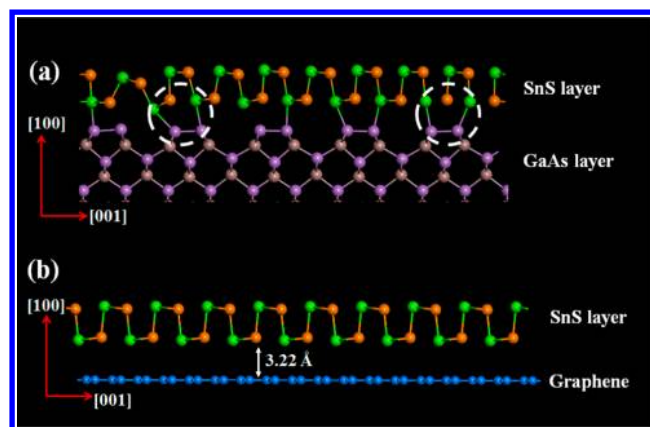


Figure 1. Optimized structures of (a) SnS/GaAs(100) and (b) SnS/graphene interfaces. Formation of chemical bonds at the SnS/GaAs interface are identified by the white circles.

$$\Delta X = \sqrt{\frac{\sum_i^n \left(\frac{b_i - b_0}{b_0} \right)^2}{n - 1}} \times 100\% \quad (1)$$

where b_i is the Sn–S bond length in the given system, b_0 is the unstrained Sn–S bond length, and n is the total number of Sn–S bonds. The bond-disorder factor, ΔX , for the SnS/GaAs(100) interface is as large as 9%, suggesting a significant lattice distortion. In contrast, ΔX for SnS/graphene interface is only 0.8%, indicating a negligible lattice distortion. The large ΔX for SnS/GaAs(100) can be attributed to the strong bonding interactions between the interfacial atoms as indicated in Figure 1a, while small ΔX for the SnS on graphene interface arises from their weak interaction as illustrated in Figure 1b. The results can be validated by the evaluation of the interfacial adhesion work, W_{ad} , of SnS on the two substrates, which is defined as $W_{ad} = (E_{SnS} + E_{Sub} - E_{tot})/A$, where E_{tot} is the total system energy and E_{SnS} and E_{sub} are the energies of the separate SnS and substrate, respectively. The calculated W_{ad} for the SnS/GaAs(100) interface is 0.52 J/m^2 , which is much larger than 0.16 J/m^2 obtained for the SnS/graphene interface. The results suggest that the significant structural disorder at the SnS/GaAs(100) interface is induced by the presence of strong chemical bonds, whereas the weak force between the SnS layer and the GBL leads to a relaxed SnS film, which ensures high crystal quality for the epilayer.

To understand the preferred growth orientations of SnS on graphene, Figure 2 illustrates the interaction energy between a graphene layer (using H-passivated C_{54} to represent graphene) and SnS as a function of their relative orientation angle where 0° is defined as SnS[100] ($a = 4.329 \text{ Å}$, $b = 11.18 \text{ Å}$, $c = 3.982 \text{ Å}$; JCPDS no. 73-1859) direction aligned along graphene $\langle 11\bar{2}0 \rangle$. In this calculation, while the SnS film follows periodic boundary conditions, the graphene layer has a finite extent. The global minimum of the surface potential energy is located at $\theta = 0^\circ$. Our calculations also indicate a local minimum at $\theta = 30^\circ$, which has a slightly higher interaction energy in the surface potential compared with $\theta = 0^\circ$. Both the global minimum and the local minimum demonstrate a periodicity of 60° due to the C_{6v} symmetry of graphene. Thus, two main growth orientations exist for SnS on graphene with the most preferred one at $\theta = i(60^\circ)$, where i is an integer.

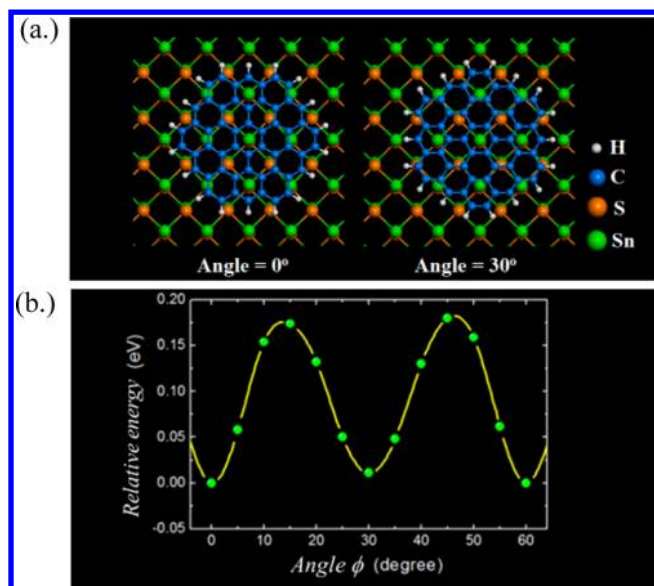


Figure 2. (a) The atomic structures for the SnS/graphene system at 0° and 30° between the SnS and graphene layers. (b) Relative surface potential energy between graphene and SnS as a function of the orientation angle ϕ .

EXPERIMENTAL RESULTS AND DISCUSSION

The theoretical calculations above are examined experimentally by the growth of orthorhombic SnS on graphene, GaAs(100), and SiO₂/Si by molecular beam epitaxy (MBE). Double layer graphene films were grown on Cu foils in a horizontal tube furnace^{26,27} at 1000 °C for 30 min with the flow of H₂/Ar gas. The graphene films were then transferred to an atomically smooth substrate (GaAs(100) or SiO₂/Si). It is observed that the quality of the SnS layers depends only on the surface roughness of the substrates and that the crystallinity of SnS/graphene/GaAs(100) and SnS/graphene/SiO₂/Si(100) are nearly the same.²⁸ The growth of the SnS thin films was performed using SnS compound source as our starting material. Typical substrate temperature was 400 °C, and the detailed experimental process is found in ref 28.

Previous studies²⁹ on the scanning electron microscopy (SEM) images of the materials indicate rough surface morphology, small grain size, and lack of a preferred orientation for SnS grown directly on GaAs(100). For SnS/graphene/GaAs, a significant improvement in the surface morphology was observed. Our results indicate significantly larger grain size with an obvious layered structure for the SnS/graphene/GaAs sample. This is in contrast to the SnS layers grown directly on GaAs(100), which exhibit much rougher surface morphology and smaller grain size.

Cross-sectional high-resolution transmission electron microscopy (HTEM) was used to investigate the interface between the SnS and the substrates. Conventional TEM sample preparation method was performed in which G1 epoxy was used to protect the sample. Due to the poor adhesion of SnS on the substrates, we have sandwiched two SnS samples between two Si wafers during the polishing and ion milling process. This technique successfully prevents the SnS film from detaching during the sample preparation process. From Figure 3a, we find that short-range order of SnS clusters is formed during the initial growth of the epilayer. It is observed that the SnS clusters nucleate in random orientations within a surrounding of amorphous materials. The result is consistent with our

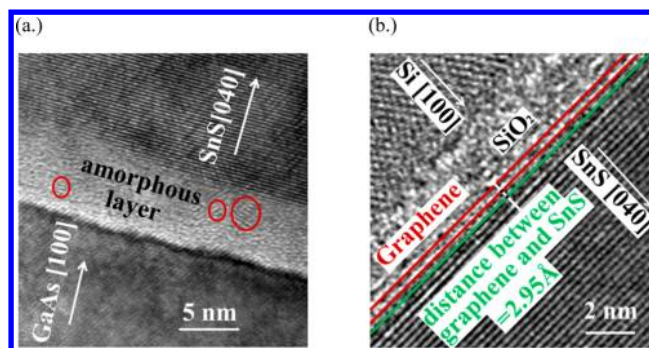


Figure 3. (a) HTEM cross-sectional image of SnS/GaAs(100) showing randomly orientated SnS clusters (red circles) surrounded by an amorphous layer during the initial growth of SnS. (b) HTEM image of SnS/graphene/SiO₂/Si showing an abrupt interface between SnS and graphene. The graphene layers are represented by red lines and the initial SnS layers are represented by the green lines.

simulation results, shown in Figure 1a, that strong chemical bonds exist between the SnS layer and the GaAs substrate. The large lattice distortion in the SnS unit layer may develop and interrupt the initial growth and result in randomly oriented SnS crystallites and amorphous layer. In contrast, Figure 3b illustrates an atomically abrupt interface between the graphene (represented by red lines) and the SnS (represented by green lines) layers. The adjacent gap between the graphene and the SnS layer is found to be 2.95 Å, which is a little smaller than the calculated value of 3.22 Å as indicated in Figure 1b suggesting a slightly stronger interaction force in reality compared with the theoretical prediction.

The experimental SEM and HTEM results are in clear agreement with the theoretical predication in which the crystallinity of the SnS layer grown on GBL is substantially improved compared with the SnS/GaAs(100) system due to the absence of strong chemical bonds at the SnS/graphene interface.

The high resolution X-ray diffraction (HXRD) θ -2 θ diffraction spectra for SnS/graphene/GaAs(100) (black line) and SnS/GaAs(100) (red line) are shown in Figure 4. Only two

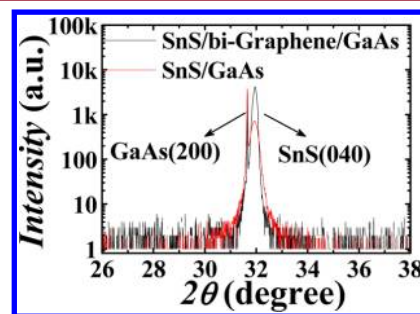


Figure 4. HXRD θ -2 θ diffraction spectra of SnS/graphene/GaAs(100) (black line) and SnS/GaAs(100) (red line).

peaks are observed at 31.9° and 66.7°, which correspond well to the (040) and (080) reflections of SnS. No other reflection peak, except for the peaks from the substrate (GaAs(200) and (400)), is observed in the scan. This indicates preferential [010] orientation for the SnS/graphene/GaAs(100) and SnS/GaAs(100) layer. The results are indicative of an orthorhombic structure with lattice constants $b_{\text{SnS/graphene}} = 11.195$ Å and $b_{\text{SnS/GaAs}} = 11.206$ Å, which agrees well with the reported values

of α -SnS ($b_{\alpha\text{-SnS}} = 11.18 \text{ \AA}$, $a_{\alpha\text{-SnS}} = 4.329 \text{ \AA}$, $c_{\alpha\text{-SnS}} = 3.982 \text{ \AA}$ (JCPDS no. 73-1859)). From Figure 4, the (040) SnS peak for the SnS/graphene/GaAs(100) layer is significantly stronger and sharper than that for the SnS/GaAs(100) layer indicating a strong preference for the [010] growth direction for the SnS/graphene/GaAs(100) film. We investigated the HXRD rocking curve measurements for the (040) plane and the fwhm for the SnS/graphene/GaAs(100) (black line) and SnS/GaAs(100) (red line) layers, which are found to be 0.37° and 2.96° , respectively, as reported in our recent publication.²⁸ The results indicate substantial improvements in the crystallinity for the SnS layer deposited on GBL. To the best of our knowledge, this is the best-reported value for SnS thin films in the literature. It is noteworthy that both Si and GaAs(100) had been used as substrates; our experimental results show that the crystallinity of the top SnS layer is a function of only the surface morphology of the substrates, and the chemical nature of the substrate has little bearing on the physical properties of the SnS layer. This clearly shows that the GBL is effective in chemically isolating the SnS epilayer from its substrates.

The nucleation of SnS on the graphene buffer layer is investigated by detailed examination of the azimuthal HXRD scans for the (160) reflection of the SnS/graphene structure. The hexagonal structure of the graphene layer yields a 6-fold symmetry for SnS grown on a single orientation of the graphene. From the inset in Figure 5, the expected 6-fold

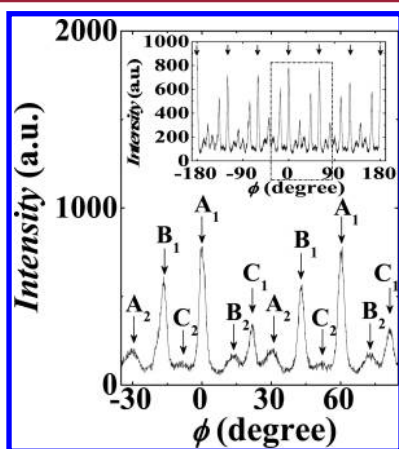


Figure 5. Azimuthal HXRD scans for (160) reflection for a typical SnS/graphene/SiO₂/Si sample.

symmetry is clearly seen (marked by arrows), which are accompanied by several additional peaks. It is important to identify the origin of these additional peaks to fully understand the growth of SnS on graphene. According to the theoretical analysis presented above, there are two preferred orientations corresponding to the global minimum at $\phi = 0^\circ$ and the local minimum at $\phi = 30^\circ$. Since both exhibit 6-fold symmetry, over a range from -30° to 85° , one expects to see two peaks corresponding to the global minimum at $\phi = 0^\circ$ and 60° (A_1) and two peaks that correspond to the local minimum at $\phi = -30^\circ$ and 30° (A_2). It is noted that, from Figure 2b above, the interaction energy for the local minimum at 30° is slightly higher than the global minimum at 0° ; thus, the XRD signals corresponding to the -30° and 30° orientations are expected to be substantially lower than those corresponding to 0° and 60° orientations. This is clearly observed from the experimental data in Figure 5. In addition, two other similar sets of XRD

peaks (B and C) are also observed. It was reported by P. Y. Huang et al.²⁹ that two or more orientations are commonly seen for graphene grown on Cu foil. Thus, the different sets of peaks observed in Figure 5 correspond to the different orientations in the graphene.

Top view HTEM measurements were performed on ultrathin SnS/graphene samples (~ 10 unit layer thick), and the results are shown in Figure 6a,b, which illustrates the grain

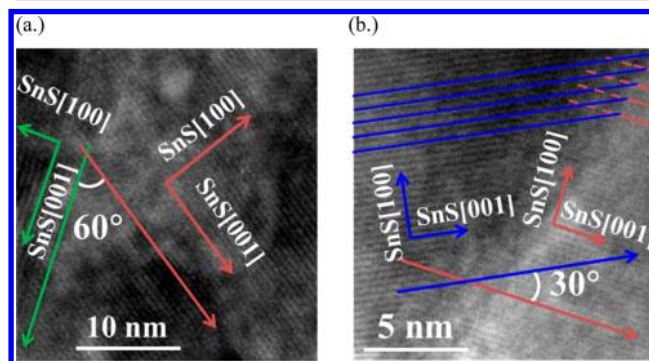


Figure 6. HTEM images of SnS clusters with orientations differing by (a) 60° and (b) 30° .

boundaries of two SnS domains oriented at 30° and 60° from each other. The observed experimental results are in excellent agreement with the theoretical prediction presented above. From the HTEM images, grain boundaries are observed when SnS clusters with different orientations merge. For SnS clusters with the same orientation, they will merge into one large cluster when they meet. However, for clusters with different orientations, grain boundaries are formed. Alternatively, one cluster may grow over another cluster of different grain orientation. While taking the HTEM pictures in Figure 6b, it was found that different grain orientations can be observed at the same spot at the grain boundary when focused at different levels. This suggests that one crystallite grows over the other, of a different orientation, at the grain boundary.

CONCLUSIONS

In conclusion, we have presented first-principle calculations to investigate the 2D/3D and 2D/2D growth of SnS. Based on our theoretical studies, chemical bonds are formed at the SnS/GaAs interface. In contrast, for SnS layers deposited on a graphene buffer layer, the interaction between the SnS layer and the graphene is dominated by vdW force. This provides the physical explanation for the substantial improvement in the crystallinity for SnS films deposited on graphene buffer layers. Our theoretical analysis on the interfacial interaction energy indicates two favored orientations for the nucleation of SnS on graphene, which are experimentally verified by detailed analyses of the HXRD and HTEM results on MBE-grown SnS layers on graphene/GaAs substrates. It is noted that the vdW interaction is weak enough so that the film is very weakly strained. Yet, it is strong enough so that the lattice orientation of the film is dictated by the substrate. This is unexpected for the completely different crystal symmetry of the film and the substrate (regular vs hexagonal). Both the weak strain and the preferred film orientations are essential for the observed excellent crystal quality.

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Notes

The authors declare no competing financial interest.

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